

Capstone Project Proposal

TSGENAI - Cohort 6

Personalized Learning Path Generator for Employee Training

1. Project Title

Personalized Learning Path Generator for Employee Training — an AI-driven, adaptive upskilling assistant that builds and continuously adjusts individualized learning journeys for every employee.

2. Brief Problem Statement

Generic, one-size-fits-all corporate training programs suffer from an average completion rate of only about 30%, because they ignore differences in employee role, prior experience, skill level, and career aspirations. Employees disengage when content is either too basic, too advanced, or irrelevant to their day-to-day work and growth goals. Organizations therefore need an intelligent system that can automatically assess each employee's current skills and goals, generate a tailored, sequenced learning path, curate the right mix of internal and external content, and adapt that path over time based on progress and feedback.

3. Background Information

3.1 Domain Information

The project sits at the intersection of Learning & Development (L&D) / HR-Tech and applied Generative AI. Enterprise L&D teams manage large, heterogeneous content libraries (internal SOPs, recorded sessions, and external MOOCs from providers such as Coursera, LinkedIn Learning, and YouTube) but lack automated, personalized mechanisms to match employees to the right content in the right order.

3.2 Problem Description and Analysis

Three structural gaps drive the low completion rates seen today:

- Lack of personalization: The same curriculum is pushed to a junior analyst and a senior manager alike, regardless of their existing competency or role trajectory.
- Static content curation: Training catalogs are not tagged or organized by skill, difficulty, or prerequisite relationships, making it hard to build a coherent sequence.
- No feedback loop: Once assigned, a learning path is rarely revisited or adjusted based on how an employee is actually performing or progressing.

An AI system combining structured skills assessment, a well-tagged content knowledge base, Retrieval-Augmented Generation (RAG) for path construction, and lightweight adaptive feedback loops can directly close these gaps.

3.3 Possible Applications

- Corporate onboarding programs for new hires across departments.
- Role-transition upskilling (e.g., Analyst → Senior Analyst → Consultant).
- Compliance and certification tracking with automated reminders.
- Cross-functional reskilling initiatives (e.g., preparing employees for AI/ML adjacent roles).
- Manager-driven development planning during performance review cycles.

4. Motivation for Selection of the Project

This project sits precisely at the intersection of Generative AI, Agentic workflows, and high-impact enterprise productivity. Unlike trivial chat-based wrappers, generating a credible 90-day learning curriculum demands:

1. Structured Knowledge Retrieval: Semantic search over heterogeneous internal/external course metadata indexed via vector embeddings.
2. Prerequisite Dependency Graphs (DAGs): Strict sequential scheduling where foundational prerequisites (e.g., Python basics & Linear Algebra) strictly precede advanced modules (e.g., Deep Learning and LangGraph orchestration).
3. Closed-Loop Adaptive Feedback (Agentic Reflection): Implementing proactive conversational agents (Slack/ Teams bots) that adjust pace, offer micro-interventions, and recalibrate syllabus parameters in response to weekly pulse checks.

5. Detailed Dataset Description and Dataset Source

Since no ready-made public dataset captures enterprise employee profiles and internal content catalogs together, the project relies primarily on synthetic data generation supplemented by publicly available content metadata.

5.1 Employee Profile Dataset (Synthetic)

A synthetic dataset of employee profiles will be generated to represent realistic organizational diversity. Fields include:

- Employee role/designation: Analyst, Senior Analyst, Consultant, Manager, etc.
- Domain of experience: past and current working experience, business domain, notable achievements/track record.
- Proficiency level per skill: Beginner, Intermediate, Expert.
- Career goals and preferred learning style (visual, hands-on, reading, etc.), collected via a Typeform/Google Forms style questionnaire schema.

5.2 Content Knowledge Base (Synthesized Spreadsheet)

A dedicated knowledge-base spreadsheet will be synthesized with the following columns, used as the retrieval corpus for the RAG pipeline:

- Skill — e.g., Python, Machine Learning, Web Development.
- Content Topic — e.g., Introduction to Python Programming, Data Science with Python, Machine Learning Fundamentals.
- Content URL — link to the corresponding course on LinkedIn Learning, Coursera, YouTube, etc.
- Metadata — difficulty level, estimated duration, format (video/reading/hands-on), and prerequisite skill tags.

5.3 Additional / Supplementary Sources

- Anonymized manual inputs from HR or training-department records (where available), used only to validate realism of the synthetic profiles.
- Publicly listed course catalogs and metadata from Coursera, LinkedIn Learning, and YouTube (titles, descriptions, durations) collected for content tagging.

All personally identifiable information will be excluded; profiles used for development and demonstration are entirely synthetic.

6. Current Benchmark

There is no single public benchmark for personalized corporate-training path generation, so the project defines its own baseline and references adjacent, well-studied problem areas:

- **Baseline:** A rule-based path generator (fixed course order per role, no personalization) will serve as the internal baseline to measure the uplift from the RAG + LLM-based approach.
- **Related work / references:** Recommender-systems literature on educational content sequencing (course-sequencing and curriculum-learning research); RAG evaluation practices from the LangChain / LlamaIndex documentation; embedding-similarity based recommendation approaches (e.g., Sentence-BERT for content similarity).
- **Industry reference points:** Adaptive learning platforms such as LinkedIn Learning's skill paths and Coursera's recommendation engine are used as qualitative benchmarks for path coherence and relevance, not as direct quantitative baselines.

7. Proposed Plan

7.1 Methodology

Approaches

- Skills assessment intake using a structured questionnaire (Typeform/Google Forms) capturing current skills, goals, and learning-style preferences, combined with manager input and historical performance data.
- Content tagging and knowledge-base construction in Notion/Airtable (and a synchronized spreadsheet) capturing skill, difficulty, duration, and format for every piece of content.
- Retrieval-Augmented Generation (RAG): given an employee profile and target skill, retrieve the most relevant tagged content and prompt an LLM (Claude/GPT) to assemble a 90-day, prerequisite-ordered learning path mixing formats.
- Adaptive feedback loop: a Slack bot collects weekly check-in responses; progress and feedback are used, together with embedding-similarity search, to recommend adjustments or next-step content.
- Gamification and tracking: progress is logged in Google Sheets, with automatic certificate generation on path completion.

Packages and Tools

- Orchestration/automation: N8N for workflow automation between forms, sheets, Slack, and the LLM API.
- Data collection: Typeform / Google Forms, Google Sheets.
- Knowledge base: Notion / Airtable for content tagging and management.
- LLM & RAG: Claude / GPT APIs, a vector store (e.g., FAISS/Chroma) for embeddings-based retrieval.
- Communication layer: Slack Bot API for adaptive check-ins and notifications.
- Prototype interface: Streamlit or Gradio for the demo dashboard; FastAPI for backend endpoints if needed.

Algorithms

- Text embeddings (e.g., OpenAI/Sentence-Transformers) for semantic similarity between employee goals, skill gaps, and content descriptions.
- RAG pipeline (retriever + LLM generator) for path synthesis and prerequisite sequencing.
- Simple graph/topological ordering to respect content prerequisite dependencies.
- Rule-based + embedding-similarity hybrid scoring for the adaptive “next step” recommendation after each check-in.

Metrics

- Path completion rate (target: 70%+).
- Time-to-proficiency / skill acquisition speed versus baseline (target: 30% faster).
- Employee satisfaction score from post-path survey (target: 85%+).
- Path relevance/coherence, evaluated qualitatively via prerequisite-ordering correctness and manager review.

7.2 Stages with Defined Deliverables

Stage	Key Activities	Deliverable
Stage 1: Data & Foundation	Generate synthetic employee profiles; build and tag content knowledge base; set up skills-assessment form.	Employee profile dataset, tagged content spreadsheet, EDA summary.
Stage 2: Path Generation Engine	Build RAG pipeline (embeddings + retrieval); design and test path-generation prompts; implement prerequisite sequencing.	Working path-generation module producing 90-day learning paths.
Stage 3: Adaptive Loop & Interface	Implement Slack bot check-ins; build embedding-similarity recommendation for next steps; build Streamlit/Gradio dashboard.	End-to-end interactive prototype with adaptive adjustment.
Stage 4: Gamification, Evaluation & Reporting	Add progress tracking and certificate generation; evaluate against baseline on defined metrics; compile final report.	Gamified tracking module, evaluation results, final report & demo.

7.3 Deployment Plan

The prototype will be deployed as a lightweight, interactive web application using Streamlit (primary) or Gradio, backed by FastAPI endpoints for the RAG and recommendation logic. This keeps the demo self-contained and easy to run locally or host on a free tier such as Hugging Face Spaces for the final showcase.

7.4 LLMOps Considerations

- Prompt versioning and logging for the path-generation and recommendation prompts.
- Basic monitoring of LLM response latency, token usage, and retrieval quality (e.g., manual spot-checks of retrieved chunks).
- Guardrails to keep generated paths within the tagged content library (avoiding hallucinated courses/URLs).
- Simple caching of embeddings and frequent RAG queries to control API cost during demos.

8. Preliminary Exploratory Data Analysis

Once the synthetic employee-profile dataset and the content knowledge-base spreadsheet are generated, preliminary EDA will cover:

- Distribution of employee roles, experience levels, and proficiency levels across the synthetic population.
- Coverage analysis of the content library — number of items per skill, per difficulty level, and per format, to identify content gaps.
- Skill-gap heatmap comparing employees' current proficiency versus target proficiency per skill.
- Basic text statistics on content descriptions (length, topic overlap) to inform embedding and chunking strategy for RAG.

9. Expected Outcomes

- A working prototype that generates a personalized, prerequisite-ordered 90-day learning path for any given synthetic employee profile.
- A functioning adaptive loop that adjusts the path based on simulated weekly progress/feedback via the Slack bot.
- A tagged content knowledge base integrated with a RAG pipeline for retrieval-grounded path generation.
- Demonstrated improvement over the rule-based baseline on completion rate, time-to-proficiency, and coherence of sequencing.
- A detailed report covering methodology, personalization logic, adaptive learning analysis, and engagement/satisfaction metrics.

10. Project Demonstration Strategy (Tentative)

- Live walkthrough of the Streamlit/Gradio dashboard: select or input an employee profile and generate its personalized learning path in real time.
- Show the RAG retrieval step explicitly (retrieved content chunks) alongside the final generated path to demonstrate grounding.
- Simulate a weekly check-in via the Slack bot to show adaptive re-sequencing of the path.
- Present the tracking dashboard (Google Sheets) showing progress and an auto-generated completion certificate.
- Close with a short metrics summary comparing the AI-generated path approach against the rule-based baseline.

11. Proposed Timeline of Project Stage Executions

The plan is organized around the three Capstone Project Mentored Sessions, with weekly progress goals as follows:

Mentored Session	Weekly Progress Goal
Session 1	Finalize problem scope; generate synthetic employee dataset; draft content-tagging schema.
Session 1	Complete content knowledge base; run preliminary EDA; set up skills-assessment form.
Session 2	Build embeddings/vector store; implement RAG retrieval over the content knowledge base.
Session 2	Develop and test path-generation prompts; implement prerequisite sequencing logic.
Session 2	Integrate Slack bot for weekly check-ins; implement embedding-similarity next-step recommendations.
Session 3	Build Streamlit/Gradio dashboard; wire up Google Sheets progress tracking and certificate generation.
Session 3	End-to-end testing; evaluate against rule-based baseline on completion rate, speed, satisfaction.
Session 3	Finalize report, prepare demonstration, polish deployment (Hugging Face Spaces / FastAPI).

12. Team Members' Names – Group 1

- Rupali Gargote
- Sarvani Medicharla

- Bowya Saran
- Kavithabharathi Natarajan
- Sahith Dhulipudi
- Bichiter Singh

13. Designated Team Coordinator's Name(s)

Team Coordinator – Rupali Gargote

Note: Replace the bracketed placeholders in Sections 12–13 with actual team details before submission.